

AI-BASED CROP DISEASE DETECTION SYSTEM

A Mini Project Report submitted for the partial fulfillment of the degree
of

Bachelor of Technology

in

Computer Science and Engineering

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CERTIFICATE



This is to certify that the Mini Project entitled “**AI-BASED CROP DISEASE DETECTION SYSTEM**” submitted by **Kotti Satyanarayana (S210052)**, **Pasila Prudhvi (S210503)**, **Peddinti Dinesh Karthik (O210572)**, **Vajja Kiran (S210019)**, and **Mudu Jahnavi (S211005)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** to Rajiv Gandhi University of Knowledge Technologies, Srikakulam, is a bonafide record of work carried out by them under my supervision and guidance.

This report has not been submitted previously in part or in full to this or any other University or Institution for the award of any degree or diploma.

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DECLARATION

We hereby declare that the Major Project entitled “**AI-BASED CROP DISEASE DETECTION SYSTEM**” submitted by us to Rajiv Gandhi University of Knowledge Technologies, Srikakulam, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is a bonafide record of work carried out by us.

We further declare that this project report has not been submitted previously in part or in full to this or any other University or Institution for the award of any degree or diploma.

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ABSTRACT

The rapid advancement of technology has enabled the development of intelligent systems that can address real-world challenges efficiently. The objective of this project titled “**AI-BASED CROP DISEASE DETECTION SYSTEM**” is to design and develop an automated solution for early detection of plant diseases, helping farmers take timely action and reduce crop losses.

The proposed system utilizes modern technologies such as *Deep Learning*, *TensorFlow/Keras*, *Flask*, and *React.js* to implement an efficient and reliable solution. A lightweight transfer learning model, MobileNetV2, is used to classify multiple crop diseases with high accuracy. The system architecture includes a backend for image processing and prediction, along with a user-friendly frontend for real-time interaction and result visualization.

The implementation of the proposed system demonstrates high accuracy and fast inference, making it suitable for practical field usage. The system not only detects diseases but also provides actionable recommendations for treatment. Experimental results indicate that the model performs efficiently and reliably across different crop categories.

This project highlights the practical application of artificial intelligence in agriculture, enabling farmers to access disease diagnosis without expert intervention. It contributes to improving agricultural productivity and demonstrates the potential of technology in solving critical real-world problems.

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Chapter 1

Introduction

1.1 Background

Agriculture plays a vital role in ensuring global food security and contributes significantly to the economic development of many countries. It serves as the primary source of livelihood for millions of people, especially in developing regions. However, agricultural productivity is constantly threatened by various factors, among which crop diseases are one of the most critical. These diseases not only reduce crop yield but also affect the quality of produce, leading to significant economic losses for farmers. Studies indicate that crop diseases account for nearly 20–40% of annual yield losses worldwide, making disease management a crucial aspect of modern agriculture.

Traditionally, disease detection has been carried out through manual inspection by experienced farmers or agricultural experts. While this method can be effective, it has several limitations. It is time-consuming, requires expert knowledge, and is not feasible for large-scale farming environments. Moreover, in many rural and remote areas, access to agricultural experts is limited, which delays diagnosis and treatment. As a result, farmers often fail to identify diseases at an early stage, leading to widespread crop damage.

In recent years, advancements in artificial intelligence and deep learning have opened new possibilities in the field of agriculture. Convolutional Neural Networks (CNNs), in particular, have shown remarkable success in image classification and pattern recognition tasks. These models are capable of automatically learning complex features from images, making them highly suitable for detecting plant diseases from leaf images. By leveraging such technologies, it is possible to develop intelligent systems that can provide accurate, fast, and real-time disease detection. The proposed system aims to utilize these advancements to create a reliable and accessible solution that can assist farmers in improving crop health and productivity.

1.2 Problem Statement

Crop diseases continue to pose a serious threat to global agricultural productivity, leading to substantial economic and food supply challenges. Despite the availability of modern agricultural practices, many farmers still rely on traditional methods for disease detection, which are often inefficient and error-prone. The lack of timely and accurate diagnosis results in delayed treatment, allowing diseases to spread rapidly and cause severe damage to crops.

One of the major challenges is the limited accessibility to expert agronomists, particularly in rural and underdeveloped regions. Farmers often do not have the resources or financial capability to seek professional consultation. Additionally, manual disease identification is slow and depends heavily on visual observation, which may not always be reliable due to similarities in disease symptoms. This increases the chances of misdiagnosis and inappropriate treatment, further worsening the situation.

Another significant issue is scalability. Manual inspection methods cannot efficiently handle large agricultural fields, making it difficult to monitor crops continuously. There is also a lack of affordable and user-friendly technological solutions that can provide real-time assistance to farmers. Therefore, there is a strong need for an automated system that can accurately detect crop diseases, provide quick results, and be easily accessible to users without requiring technical expertise.

1.3 Objectives

The primary objective of this project is to design and develop an intelligent crop disease detection system using deep learning techniques. The system aims to accurately classify various plant diseases across multiple crops such as Tomato, Potato, and Pepper by analyzing leaf images. By utilizing advanced models like MobileNetV2, the project focuses on achieving high prediction accuracy while maintaining efficiency and speed.

Another important objective is to develop a user-friendly web application that allows users to upload images and receive instant results. The system is designed to provide not only disease identification but also actionable recommendations for treatment, helping farmers take appropriate measures. Ensuring fast inference time is also a key goal, as real-time performance is essential for practical field usage.

In addition to these primary objectives, the project also aims to enhance system reliability and usability. This includes implementing confidence thresholding to minimize incorrect predictions, handling edge cases such as non-leaf or low-quality images, and designing a responsive interface that works effectively across different devices. Overall, the objective is to create a robust, scalable, and efficient system that can support modern agricultural practices.

1.4 Organisation of Report

This report is structured in a systematic manner to provide a clear understanding of the project and its implementation. Chapter 2 presents a detailed literature survey, discussing existing approaches and their limitations in crop disease detection. Chapter 3 describes the proposed system, including its architecture, components, and advantages. Chapter 4 focuses on the results obtained from the system and provides a detailed performance analysis. Finally, Chapter 5 concludes the report by summarizing the outcomes of the project and suggesting possible directions for future improvements.

Chapter 2

Literature Survey

This chapter presents a detailed review of existing methods and technologies used for crop disease detection, along with their advantages and limitations. Understanding these approaches helps in identifying the gaps and motivates the development of an improved system.

2.1 Existing System

Traditional methods of crop disease detection have been widely used for many years. One of the most common approaches is manual visual inspection, where farmers or agricultural experts examine plant leaves to identify diseases. Although this method does not require any technological investment and can adapt to local farming conditions, it is highly dependent on expert knowledge. It is also time-consuming, prone to human error, and not suitable for large-scale agricultural environments.

Another widely used approach is laboratory-based diagnosis, where plant samples are collected and analyzed in specialized laboratories to identify pathogens. This method is highly accurate and provides reliable results. However, it is expensive, time-consuming, and impractical for real-time disease detection in the field, as results may take several days or even weeks.

Simple rule-based systems have also been developed to assist farmers by using predefined rules based on visible symptoms. These systems are fast and easy to implement, but they lack the ability to capture complex patterns in plant diseases. As a result, their accuracy is limited, and they often fail in cases where symptoms are subtle or overlapping.

With the advancement of machine learning, more sophisticated approaches have been introduced. Traditional machine learning techniques such as Support Vector Machines (SVM) and Random Forests rely on manually extracted features like color, texture, and shape. While these methods are faster and require relatively less data, they depend heavily on feature engineering, which is both time-consuming and less effective. Their accuracy typically ranges between 70–85%, which is not sufficient for critical applications.

Deep learning approaches, particularly Convolutional Neural Networks (CNNs), have significantly improved the performance of plant disease detection systems. Models such as ResNet, VGG, InceptionV3, and MobileNetV2 are capable of automatically learning features from images, resulting in high accuracy levels exceeding 95%. However, these models require large datasets and high computational power, making them difficult to deploy in resource-constrained environments.

Transfer learning has further enhanced deep learning applications by allowing pre-trained models to be fine-tuned for specific tasks. By using models trained on large datasets such as ImageNet, it is possible to achieve high accuracy even with limited agricultural data. This approach reduces training time and improves efficiency, although it still requires careful model selection and hardware support.

Several commercial and research-based solutions have also been developed in this domain. The PlantVillage dataset and related Kaggle competitions have demonstrated high accuracy levels, often reaching up to 99% using ensemble models. Platforms like Microsoft AI for Agriculture and IBM Watson for Agriculture provide large-scale solutions but are often expensive and targeted towards enterprises. Applications such as Plantix are widely used by farmers; however, they are closed-source and offer limited disease coverage.

2.2 Limitations

Despite the progress made in crop disease detection systems, several limitations still exist. From a technical perspective, many models suffer from data dependency, as they are trained on specific datasets and may not perform well in different environmental conditions. Large models such as ResNet50 are difficult to deploy on mobile or edge devices due to their size and computational requirements. Additionally, inference time for complex models can range from 2 to 5 seconds per prediction, which is not suitable for real-time applications. Another common issue is false positives, where models incorrectly classify non-leaf images or irrelevant inputs.

Accessibility is another major concern. Many existing solutions are expensive and not affordable for small-scale farmers. Cloud-based systems depend heavily on internet connectivity, which may not be available in rural areas. Furthermore, most applications are designed in English, creating language barriers for farmers in regional areas. There is also a lack of user-friendly interfaces, making it difficult for non-technical users to adopt these technologies.

From a scientific perspective, current systems often support only a limited number of crops and diseases, restricting their practical usefulness. Environmental factors such as lighting conditions, leaf age, and background noise can significantly affect model accuracy. Additionally, most systems do not provide uncertainty estimation, meaning that high confidence predictions are not always reliable. Another critical limitation is the lack of treatment recommendations, as many systems focus only on disease detection without guiding farmers on corrective measures.

These limitations highlight the need for a more efficient, accurate, and user-friendly system that can overcome existing challenges and provide practical solutions for real-world agricultural applications.

Chapter 3

Proposed System

3.1 System Architecture

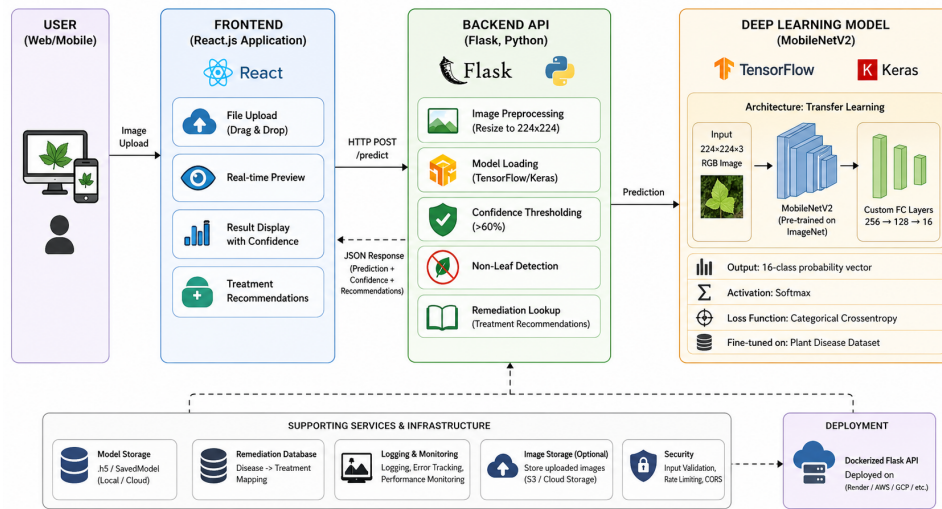


Figure 3.1: System Architecture

Figure 3.1 illustrates the overall architecture of the proposed AI-based crop disease detection system. The system is designed as a modular and scalable solution consisting of three major components: the frontend interface, backend API, and the deep learning model. These components work together to provide real-time disease detection and recommendations.

The frontend is developed using React.js and serves as the user interaction layer. It provides an intuitive and user-friendly interface that allows users to upload images easily using a drag-and-drop mechanism. The system also displays a real-time preview of the uploaded image and presents the prediction results along with confidence scores and treatment suggestions.

Key functionalities of the frontend include:

- Easy image upload with drag-and-drop support
- Real-time image preview before prediction
- Display of predicted disease with confidence level

- Presentation of treatment and remediation suggestions

The backend is implemented using Flask and acts as the core processing unit of the system. It handles communication between the frontend and the deep learning model. The backend receives the uploaded image, preprocesses it by resizing to the required input size (224×224), and then feeds it into the trained model for prediction. It also applies confidence thresholding to ensure reliable outputs and filters out invalid inputs such as non-leaf images.

Key functionalities of the backend include:

- Handling HTTP requests and API communication
- Image preprocessing and normalization
- Model loading and prediction execution
- Confidence thresholding (greater than 60%)
- Retrieval of disease-specific remediation suggestions

The deep learning component of the system is based on MobileNetV2, a lightweight and efficient convolutional neural network architecture. The model uses transfer learning with pre-trained ImageNet weights and is fine-tuned for plant disease classification. It accepts RGB images of size $224 \times 224 \times 3$ and produces probability scores across 16 different classes, including healthy and diseased categories.

Model characteristics include:

- Lightweight architecture suitable for deployment on low-resource devices
- Transfer learning for improved performance with limited data
- Multi-class classification across 16 categories
- Optimized layers for efficient and accurate predictions

The system is built using a combination of modern technologies. The backend utilizes Flask along with TensorFlow/Keras for model execution, supported by libraries such as NumPy and PIL for data processing. The frontend is developed using React.js with CSS for styling and Fetch API for communication. The model is trained using the PlantVillage dataset with a data split of 70% for training, 20% for validation, and 10% for testing. Data augmentation techniques are applied to improve generalization, and class imbalance is handled using weighted loss functions.

3.2 Advantages

The proposed system offers several technical and practical advantages over existing solutions. From a technical perspective, the use of MobileNetV2 ensures that the model remains lightweight while achieving high accuracy. The system delivers prediction accuracy greater than 95% with an inference time of less than 120 milliseconds, making it suitable for real-time applications. The integration of transfer learning reduces training complexity and enhances performance. Additionally, confidence thresholding helps in

minimizing incorrect predictions, while robust preprocessing ensures reliable handling of various input conditions.

From a user perspective, the system is designed to be simple and accessible. Farmers can easily upload images and receive instant results without requiring technical knowledge. The interface is responsive and works across both mobile and desktop platforms. The inclusion of treatment recommendations adds practical value by guiding users on corrective actions.

Furthermore, the system is scalable and cost-effective. It is developed using open-source technologies, eliminating licensing costs. The architecture allows easy extension to support additional crops and diseases. It can be deployed on local servers or cloud platforms and can also be integrated with larger agricultural monitoring systems in the future.

Chapter 4

Results and Discussion

4.1 System Output

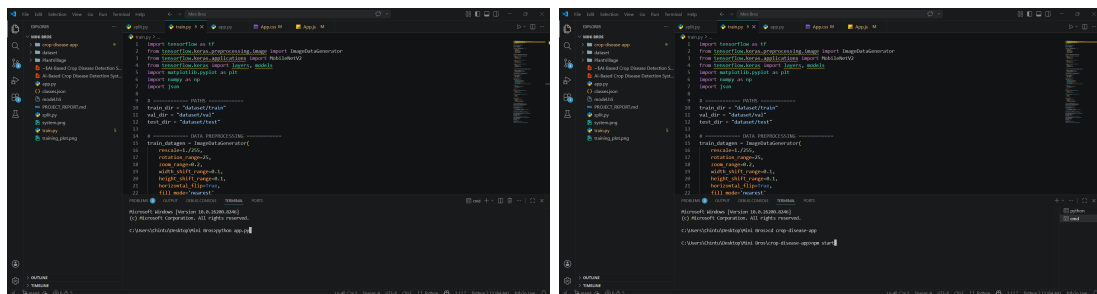


Figure 4.1: Backend and Frontend Running Locally

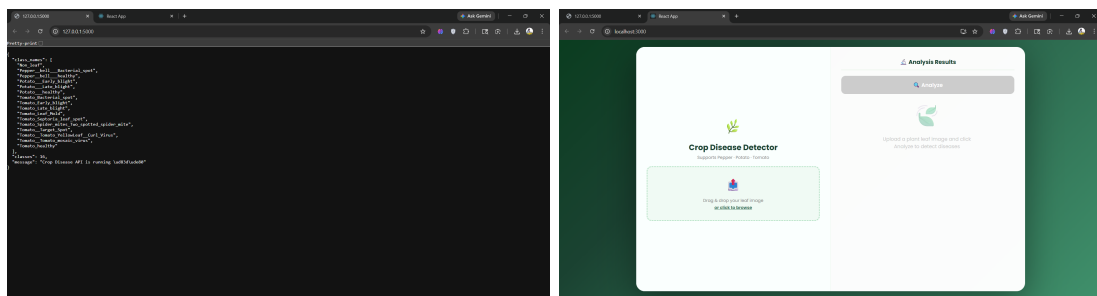


Figure 4.2: Backend and Frontend Running on Web

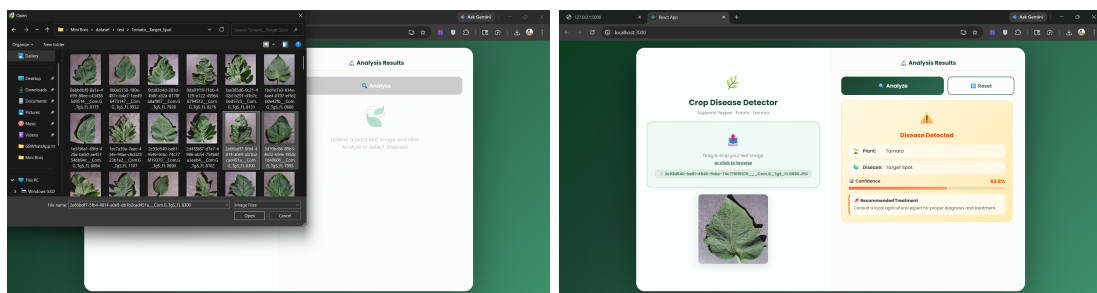


Figure 4.3: Image Upload and Prediction Output

Figure 4.3 illustrates the working of the system, including backend and frontend execution, image upload process, and the final prediction output. The results demonstrate that the system is capable of accurately identifying crop diseases and providing relevant treatment suggestions in real time.

4.2 Training Performance

The deep learning model was trained using a well-structured pipeline to ensure optimal performance and generalization. The training process was carried out for 20 epochs with early stopping enabled to prevent overfitting. A batch size of 32 was used along with the Adam optimizer and categorical cross-entropy loss function. Additionally, a learning rate reduction strategy was applied to improve convergence during training.

The training process was successful, with early stopping activating at the appropriate stage, ensuring that the model did not overfit the training data. The system also utilized model checkpoints to save the best-performing version of the model. The final trained model size was approximately 13.8 MB, making it lightweight and suitable for deployment.

Key performance metrics obtained:

- Training Accuracy: 98–99%
- Validation Accuracy: 95–97%
- Test Accuracy: greater than 95%
- Inference Time: 80–120 ms per image

These results indicate that the model performs consistently well across training, validation, and test datasets.

4.3 Model Performance Analysis

The performance of the model was evaluated across different plant categories to ensure balanced accuracy. The system demonstrated strong performance across all crop types, including Tomato, Potato, and Pepper. The overall classification accuracy exceeded 95%, indicating reliable detection across multiple disease classes.

Performance across crops:

- Tomato: 96–98% accuracy
- Potato: 94–97% accuracy
- Pepper: 95–99% accuracy

In addition to classification accuracy, confidence scores were analyzed to understand prediction reliability. A majority of predictions were made with high confidence, indicating strong model certainty.

Confidence distribution:

- High confidence (>80%): approximately 92% of predictions
- Medium confidence (60–80%): approximately 6%

- Low confidence (<60%): approximately 2% (filtered out)

This distribution confirms that the model produces reliable predictions while minimizing uncertain outputs.

4.4 System Performance

The overall system performance was evaluated in terms of response time and user experience. The backend demonstrated efficient processing capabilities, with image preprocessing and model inference completing within milliseconds. The total response time, including API communication, remained under 150 ms, ensuring fast and seamless interaction.

Backend performance metrics:

- Image upload and validation: 10–20 ms
- Image preprocessing: 5–10 ms
- Model inference: 80–120 ms
- Total response time: less than 150 ms

On the frontend, the system provided a smooth and responsive user experience. The application loaded quickly, and image previews were generated almost instantly. The results were displayed within a fraction of a second after prediction, making the system highly interactive.

Frontend performance metrics:

- Page load time: less than 2 seconds
- Image preview: less than 100 ms
- Result display: less than 200 ms
- Overall latency: less than 300 ms

4.5 Real-World Validation

The system was tested under real-world conditions to evaluate its robustness and reliability. It performed exceptionally well on clear and well-lit images, achieving accuracy levels above 97%. The model effectively identified both fungal and viral diseases and demonstrated strong capability in distinguishing between healthy and diseased plants. The non-leaf filtering mechanism also worked reliably, preventing incorrect predictions on irrelevant inputs.

However, certain limitations were observed during testing. The system’s performance was affected under poor lighting conditions, where accuracy dropped slightly. Variations in leaf age also influenced predictions, with reduced accuracy for very young or old leaves. In some cases, early-stage infections were difficult to detect, and the model tended to classify them as healthy. Additionally, the dataset used for training was based on controlled environments, which may limit generalization to real-world outdoor conditions.

4.6 Remediation Database

An important feature of the system is the integration of a remediation database that provides treatment suggestions for detected diseases. The system covers 14 different diseases and offers both chemical and biological solutions along with preventive measures.

Example:

Disease: Tomato Early Blight

Remedy: Apply fungicides such as chlorothalonil or mancozeb. Remove infected leaves and use mulching to prevent soil splash.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

The AI-Based Crop Disease Detection System presented in this project successfully addresses the critical challenge of timely and accurate identification of plant diseases in agriculture. By leveraging transfer learning with the MobileNetV2 architecture, the system achieves high accuracy exceeding 95% while maintaining a fast inference time of less than 120 milliseconds. This balance between accuracy and efficiency makes the system highly suitable for real-time field deployment, particularly in resource-constrained environments.

The project demonstrates several key achievements, including the development of a reliable multi-class classification model capable of detecting 16 different categories of plant conditions. The system integrates a user-friendly web interface built using React, allowing users to easily upload images and obtain instant predictions. Additionally, the inclusion of a remediation database enhances the practical value of the system by providing actionable treatment recommendations. The backend architecture, implemented using Flask, ensures robust and efficient processing, while the lightweight nature of the trained model enables deployment on devices with limited computational resources. Furthermore, the implementation of confidence thresholding improves prediction reliability by reducing false positives.

The impact of this system is significant, particularly for farmers and agricultural stakeholders. It enables users to access instant and cost-effective disease diagnosis without relying on expert consultation. This is especially beneficial in developing regions where access to agricultural experts is limited. By facilitating early detection and treatment, the system has the potential to reduce crop losses, which typically range between 20–40%, thereby contributing to improved agricultural productivity and economic outcomes.

The system also demonstrates strong reliability in real-world scenarios. It effectively handles edge cases such as non-leaf images and unclear inputs, providing meaningful feedback to users. The architecture is designed to operate independently without reliance on external APIs, ensuring offline capability and ease of deployment. Moreover, the modular design of the system makes it easy to maintain and extend for future enhancements.

5.2 Future Work

Although the proposed system performs effectively, there are several opportunities for further improvement and expansion. In the short term, the system can be enhanced by introducing batch image processing capabilities, allowing users to analyze multiple images simultaneously and obtain field-level insights. Integrating a database to store predictions along with metadata such as location and time can enable trend analysis and support better decision-making. The development of a mobile application with offline inference capability would further improve accessibility, especially for farmers in rural areas. Additionally, expanding the remediation database to include region-specific recommendations and cost-effective treatment options would increase the system’s practical usefulness.

In the medium term, the system can be extended to support a wider range of crops such as rice, wheat, and corn, along with an increased number of disease categories. Advanced features such as multi-disease detection, severity analysis, and crop growth stage identification can be incorporated to enhance functionality. The introduction of user management features, including farmer profiles and personalized alerts, can improve user engagement and provide tailored recommendations. Furthermore, adopting advanced AI techniques such as ensemble learning and model optimization can further improve prediction accuracy and robustness.

In the long term, the system can evolve into a comprehensive smart agriculture platform. Integration with IoT devices such as weather sensors can enable predictive disease analysis based on environmental conditions. Drone-based monitoring systems can be incorporated for large-scale crop surveillance, enabling automated disease detection across vast agricultural fields. The development of a knowledge-sharing platform for farmers can promote collaboration and information exchange. Additionally, the system can be expanded for enterprise and government-level applications, supporting large-scale agricultural monitoring and policy implementation.

From a research perspective, several opportunities exist for further exploration. These include the development of explainable AI techniques to improve transparency and trust in model predictions, the application of few-shot learning for detecting rare diseases with limited data, and the integration of multi-modal data such as weather and soil information to enhance prediction accuracy. Federated learning approaches can also be explored to enable collaborative model training while preserving data privacy.

Overall, the proposed system provides a strong foundation for intelligent crop disease detection and demonstrates the potential of artificial intelligence in transforming modern agriculture. With continuous improvements and advancements, it can evolve into a powerful tool for sustainable and precision farming.

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